

AuraHealth HMS

Hospital Management System Dashboard

Report: Daily Codebase Update

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Daily Updates Report

This report details the updates committed to the Hospital Management System workspace today. The implementations focus on adding standalone desktop capabilities and offline data robustness to the client dashboard interface.

KEY ACCOMPLISHMENTS

1. Desktop Application Wrapper (Electron Integration)

Implemented a package infrastructure allowing operators to launch the AuraHealth console inside an isolated desktop shell. The environment includes lifecycle hooks, custom application icons, and dynamic installer configuration.

2. Offline-First IndexedDB Sync Engine

Integrated a robust client data caching scheme utilizing IndexedDB. Write actions performed during low or disconnected network status are deferred to an offline log queue, while read requests are answered locally. The engine performs chronological updates and notifies users of synchronization states once server connectivity resumes.

TECHNICAL IMPLEMENTATION BREAKDOWN

Desktop Interface (Electron Wrapper)

- **Window Lifecycle Controller:** Added `main.cjs` targeting security boundaries, window configuration (1440x900), and dev console behavior.
- **Security Middleware:** Implemented `preload.cjs` for secure context isolation and script containment.
- **Admin Shortcuts:** Crafted `createShortcut.ps1` enabling automated shortcut installation mapped to custom brand assets.
- **Scripts Pipeline:** Augmented `package.json` with scripts to trigger the electron dev interface.

Offline-First Network Sync Service

- **Local Database Scheme:** Crafted `localDb.js` establishing database versions and collection object stores (doctors, patients, medicines, offlineQueue).
- **Synchronization Engine:** Developed `syncService.js`, managing network states, dispatching queued actions, replacing temporary client IDs, and initiating custom toast UI alerts.
- **HTTP Client Middleware:** Patched `client.js` to intercept connection loss, seed databases, and fall back to local IndexedDB writes.

- **Integrated UI Indicators:** Patched `Header.jsx` and `ToastContext.jsx` to bind synchronized notifications and state indicators into the user dashboard layout.

FILE MANIFEST

Status	File Path	Description
NEW FILE	<code>hms-frontend/main.cjs</code>	Electron main process script defining boundaries, icon, and dev tools.
NEW FILE	<code>hms-frontend/preload.cjs</code>	Secure bridge layer between desktop shell and React frontend.
NEW FILE	<code>hms-frontend/createShortcut.ps1</code>	PowerShell installer mapping Desktop shortcuts to console launcher.
NEW FILE	<code>hms-frontend/src/utils/localDb.js</code>	IndexedDB interface to cache profiles and manage mutations offline.
NEW FILE	<code>hms-frontend/src/utils/syncService.js</code>	Background event engine handling sync retries and conflict toast alerts.
NEW FILE	<code>hms-frontend/public/logo.ico</code>	System desktop display icon asset.
NEW FILE	<code>hms-frontend/public/logo.png</code>	High resolution web logo asset.
MODIFIED	<code>hms-frontend/package.json</code>	Added electron dependencies and package start commands.
MODIFIED	<code>hms-frontend/src/api/client.js</code>	API client hook for database caching and offline network interception.
MODIFIED	<code>hms-frontend/src/components/Layout/Header.jsx</code>	Injected network connection state badge and sync icons.
MODIFIED	<code>hms-frontend/src/context/ToastContext.jsx</code>	Wired custom event listeners to render synchronization toasts.